Assignment 3.1 Exercise

In this assignment, you will explore the impact of fairness metrics after applying the Disparate Impact Remover (DIR), retrain a Light GBM model with the adjusted data, and compare its metrics (listed below) against the original and reweighted models from the lab tutorial. Afterward, respond to the reflection questions.

Part 1: Experimentation with the DIR

- Modify the levels array for the Disparate Impact Remover to include finer and coarser granularity. (See example in lab/assignment notebook.)
- Identify the best repair level and the Disparate Impact at the best repair level.
- Retrain a Light GBM model on the modified data from the DIR and compute both the performance and fairness metrics.

1.1a Code Output for Finer Granularity

Paste your screenshot here. (This can be a very lengthy output)

Attach a screenshot or output of your code here:

```
  1%|          | 1/101 [00:11<18:36, 11.17s/it]Repair Level: 0.00, Disparate Impact: 0.8350
  2%||          | 2/101 [00:18<15:06,  9.16s/it]Repair Level: 0.00, Disparate Impact: 0.8434
  3%||          | 3/101 [00:28<15:31,  9.50s/it]Repair Level: 0.00, Disparate Impact: 0.8464
  4%||          | 4/101 [00:38<15:29,  9.59s/it]Repair Level: 0.00, Disparate Impact: 0.8467
  5%||          | 5/101 [00:46<14:10,  8.86s/it]Repair Level: 0.01, Disparate Impact: 0.8401
  6%|▍          | 6/101 [00:56<14:58,  9.45s/it]Repair Level: 0.01, Disparate Impact: 0.8448
  7%|▍          | 7/101 [01:06<14:43,  9.40s/it]Repair Level: 0.01, Disparate Impact: 0.8405
  8%|▍          | 8/101 [01:15<14:23,  9.28s/it]Repair Level: 0.01, Disparate Impact: 0.8367
  9%|▍          | 9/101 [01:25<14:45,  9.63s/it]Repair Level: 0.01, Disparate Impact: 0.8419
 10%|▍          | 10/101 [01:35<14:51,  9.79s/it]Repair Level: 0.01, Disparate Impact: 0.8456
 11%|▍          | 11/101 [01:44<14:08,  9.43s/it]Repair Level: 0.01, Disparate Impact: 0.8330
 12%|▍          | 12/101 [01:54<14:26,  9.74s/it]Repair Level: 0.01, Disparate Impact: 0.8420
 13%|▍          | 13/101 [02:04<14:27,  9.86s/it]Repair Level: 0.01, Disparate Impact: 0.8332
 14%|▍          | 14/101 [02:13<13:46,  9.50s/it]Repair Level: 0.02, Disparate Impact: 0.8411
 15%|▍          | 15/101 [02:24<14:07,  9.86s/it]Repair Level: 0.02, Disparate Impact: 0.8338
 16%|▍          | 16/101 [02:32<13:19,  9.40s/it]Repair Level: 0.02, Disparate Impact: 0.8443
 17%|▍          | 17/101 [02:41<13:02,  9.31s/it]Repair Level: 0.02, Disparate Impact: 0.8365
 18%|▍          | 18/101 [02:52<13:39,  9.87s/it]Repair Level: 0.02, Disparate Impact: 0.8367
 19%|▍          | 19/101 [03:00<12:40,  9.27s/it]Repair Level: 0.02, Disparate Impact: 0.8393
 20%|▍          | 20/101 [03:10<12:37,  9.36s/it]Repair Level: 0.02, Disparate Impact: 0.8381
 21%|▍          | 21/101 [03:19<12:38,  9.48s/it]Repair Level: 0.03, Disparate Impact: 0.8398
 22%|▍          | 22/101 [03:28<11:56,  9.07s/it]Repair Level: 0.03, Disparate Impact: 0.8440
 23%|▍          | 23/101 [03:38<12:12,  9.40s/it]Repair Level: 0.03, Disparate Impact: 0.8317
 24%|▍          | 24/101 [03:46<11:25,  8.91s/it]Repair Level: 0.03, Disparate Impact: 0.8336
 25%|▍          | 25/101 [03:55<11:25,  9.02s/it]Repair Level: 0.03, Disparate Impact: 0.8422
 26%|▍          | 26/101 [04:05<11:47,  9.44s/it]Repair Level: 0.03, Disparate Impact: 0.8390
 27%|▍          | 27/101 [04:13<11:02,  8.95s/it]Repair Level: 0.03, Disparate Impact: 0.8403
 28%|▍          | 28/101 [04:23<11:08,  9.16s/it]Repair Level: 0.03, Disparate Impact: 0.8367
 29%|▍          | 29/101 [04:32<11:09,  9.29s/it]Repair Level: 0.04, Disparate Impact: 0.8363
 30%|▍          | 30/101 [04:40<10:31,  8.90s/it]Repair Level: 0.04, Disparate Impact: 0.8459
 31%|▍          | 31/101 [04:51<11:02,  9.47s/it]Repair Level: 0.04, Disparate Impact: 0.8439
 32%|▍          | 32/101 [05:00<10:35,  9.21s/it]Repair Level: 0.04, Disparate Impact: 0.8439
 33%|▍          | 33/101 [05:09<10:28,  9.25s/it]Repair Level: 0.04, Disparate Impact: 0.8417
 34%|▍          | 34/101 [05:20<10:53,  9.75s/it]Repair Level: 0.04, Disparate Impact: 0.8429
 35%|▍          | 35/101 [05:28<10:08,  9.21s/it]Repair Level: 0.04, Disparate Impact: 0.8447
 36%|▍          | 36/101 [05:37<10:04,  9.30s/it]Repair Level: 0.04, Disparate Impact: 0.8400
 37%|▍          | 37/101 [05:47<10:04,  9.44s/it]Repair Level: 0.04, Disparate Impact: 0.8406
 38%|▍          | 38/101 [05:56<09:36,  9.15s/it]Repair Level: 0.05, Disparate Impact: 0.8307
 39%|▍          | 39/101 [06:05<09:34,  9.27s/it]Repair Level: 0.05, Disparate Impact: 0.8439
 40%|▍          | 40/101 [06:14<09:13,  9.07s/it]Repair Level: 0.05, Disparate Impact: 0.8368
 41%|▍          | 41/101 [06:22<08:55,  8.93s/it]Repair Level: 0.05, Disparate Impact: 0.8377
 42%|▍          | 42/101 [06:33<09:08,  9.30s/it]Repair Level: 0.05, Disparate Impact: 0.8338
 43%|▍          | 43/101 [06:41<08:42,  9.02s/it]Repair Level: 0.05, Disparate Impact: 0.8327
 44%|▍          | 44/101 [06:50<08:35,  9.05s/it]Repair Level: 0.05, Disparate Impact: 0.8366
 45%|▍          | 45/101 [07:00<08:35,  9.21s/it]Repair Level: 0.06, Disparate Impact: 0.8407
 46%|▍          | 46/101 [07:07<08:04,  8.80s/it]Repair Level: 0.06, Disparate Impact: 0.8358
 47%|▍          | 47/101 [07:17<08:09,  9.07s/it]Repair Level: 0.06, Disparate Impact: 0.8362
 48%|▍          | 48/101 [07:25<07:41,  8.71s/it]Repair Level: 0.06, Disparate Impact: 0.8370
 49%|▍          | 49/101 [07:35<07:48,  9.01s/it]Repair Level: 0.06, Disparate Impact: 0.8369
 50%|▍          | 50/101 [07:44<07:49,  9.21s/it]Repair Level: 0.06, Disparate Impact: 0.8404
 50%|▍          | 51/101 [07:52<07:18,  8.76s/it]Repair Level: 0.06, Disparate Impact: 0.8259
 51%|▍          | 52/101 [08:02<07:29,  9.18s/it]Repair Level: 0.06, Disparate Impact: 0.8374
 52%|▍          | 53/101 [08:11<07:21,  9.19s/it]Repair Level: 0.07, Disparate Impact: 0.8339
 53%|▍          | 54/101 [08:20<06:59,  8.92s/it]Repair Level: 0.07, Disparate Impact: 0.8376
 54%|▍          | 55/101 [08:30<07:05,  9.25s/it]Repair Level: 0.07, Disparate Impact: 0.8403
 55%|▍          | 56/101 [08:38<06:37,  8.84s/it]Repair Level: 0.07, Disparate Impact: 0.8370
 56%|▍          | 57/101 [08:48<06:42,  9.14s/it]Repair Level: 0.07, Disparate Impact: 0.8337
 57%|▍          | 58/101 [08:58<06:47,  9.47s/it]Repair Level: 0.07, Disparate Impact: 0.8298
 58%|▍          | 59/101 [09:06<06:24,  9.16s/it]Repair Level: 0.07, Disparate Impact: 0.8325
 59%|▍          | 60/101 [09:17<06:30,  9.53s/it]Repair Level: 0.07, Disparate Impact: 0.8372
 60%|▍          | 61/101 [09:26<06:20,  9.52s/it]Repair Level: 0.07, Disparate Impact: 0.8327
 61%|▍          | 62/101 [09:34<05:56,  9.14s/it]Repair Level: 0.08, Disparate Impact: 0.8372
 62%|▍          | 63/101 [09:45<05:59,  9.46s/it]Repair Level: 0.08, Disparate Impact: 0.8477
 63%|▍          | 64/101 [09:54<05:45,  9.33s/it]Repair Level: 0.08, Disparate Impact: 0.8378
 64%|▍          | 65/101 [10:03<05:33,  9.27s/it]Repair Level: 0.08, Disparate Impact: 0.8339
 65%|▍          | 66/101 [10:13<05:37,  9.65s/it]Repair Level: 0.08, Disparate Impact: 0.8383
 66%|▍          | 67/101 [10:22<05:15,  9.27s/it]Repair Level: 0.08, Disparate Impact: 0.8285
 67%|▍          | 68/101 [10:31<05:06,  9.29s/it]Repair Level: 0.08, Disparate Impact: 0.8376
 68%|▍          | 69/101 [10:41<05:06,  9.58s/it]Repair Level: 0.09, Disparate Impact: 0.8313
 69%|▍          | 70/101 [10:49<04:44,  9.18s/it]Repair Level: 0.09, Disparate Impact: 0.8313
 70%|▍          | 71/101 [10:59<04:36,  9.22s/it]Repair Level: 0.09, Disparate Impact: 0.8284
 71%|▍          | 72/101 [11:08<04:24,  9.12s/it]Repair Level: 0.09, Disparate Impact: 0.8357
 72%|▍          | 73/101 [11:16<04:07,  8.84s/it]Repair Level: 0.09, Disparate Impact: 0.8402
 73%|▍          | 74/101 [11:26<04:07,  9.17s/it]Repair Level: 0.09, Disparate Impact: 0.8389
 74%|▍          | 75/101 [11:34<03:49,  8.84s/it]Repair Level: 0.09, Disparate Impact: 0.8428
 75%|▍          | 76/101 [11:44<03:49,  9.18s/it]Repair Level: 0.09, Disparate Impact: 0.8341
 76%|▍          | 77/101 [11:54<03:46,  9.45s/it]Repair Level: 0.10, Disparate Impact: 0.8385
 77%|▍          | 78/101 [12:02<03:27,  9.04s/it]Repair Level: 0.10, Disparate Impact: 0.8407
 78%|▍          | 79/101 [12:12<03:24,  9.28s/it]Repair Level: 0.10, Disparate Impact: 0.8421
 79%|▍          | 80/101 [12:21<03:16,  9.36s/it]Repair Level: 0.10, Disparate Impact: 0.8347
 80%|▍          | 81/101 [12:30<03:01,  9.07s/it]Repair Level: 0.10, Disparate Impact: 0.8421
 81%|▍          | 82/101 [12:41<03:02,  9.60s/it]Repair Level: 0.20, Disparate Impact: 0.8495
 82%|▍          | 83/101 [12:50<02:51,  9.53s/it]Repair Level: 0.24, Disparate Impact: 0.8494
 83%|▍          | 84/101 [12:59<02:37,  9.24s/it]Repair Level: 0.28, Disparate Impact: 0.8433
 84%|▍          | 85/101 [13:09<02:33,  9.57s/it]Repair Level: 0.33, Disparate Impact: 0.8539
 85%|▍          | 86/101 [13:17<02:15,  9.06s/it]Repair Level: 0.37, Disparate Impact: 0.8566
 86%|▍          | 87/101 [13:26<02:09,  9.22s/it]Repair Level: 0.41, Disparate Impact: 0.8597
 87%|▍          | 88/101 [13:35<01:58,  9.14s/it]Repair Level: 0.45, Disparate Impact: 0.8605
 88%|▍          | 89/101 [13:44<01:47,  8.96s/it]Repair Level: 0.49, Disparate Impact: 0.8659
 89%|▍          | 90/101 [13:54<01:42,  9.29s/it]Repair Level: 0.54, Disparate Impact: 0.7930
 90%|▍          | 91/101 [14:01<01:26,  8.67s/it]Repair Level: 0.58, Disparate Impact: 0.7894
 91%|▍          | 92/101 [14:10<01:19,  8.81s/it]Repair Level: 0.62, Disparate Impact: 0.7959
 92%|▍          | 93/101 [14:20<01:11,  8.98s/it]Repair Level: 0.66, Disparate Impact: 0.8057
 93%|▍          | 94/101 [14:28<01:00,  8.68s/it]Repair Level: 0.71, Disparate Impact: 0.8161
 94%|▍          | 95/101 [14:37<00:53,  8.92s/it]Repair Level: 0.75, Disparate Impact: 0.8368
 95%|▍          | 96/101 [14:45<00:42,  8.48s/it]Repair Level: 0.79, Disparate Impact: 0.8335
 96%|▍          | 97/101 [14:54<00:34,  8.66s/it]Repair Level: 0.83, Disparate Impact: 0.8341
 97%|▍          | 98/101 [15:01<00:25,  8.41s/it]Repair Level: 0.87, Disparate Impact: 0.8257
 98%|▍          | 99/101 [15:10<00:16,  8.36s/it]Repair Level: 0.92, Disparate Impact: 0.8142
 99%|▍          | 100/101 [15:18<00:08,  8.39s/it]Repair Level: 0.96, Disparate Impact: 0.8074
100%|▍          | 101/101 [15:26<00:00,  9.17s/it]Repair Level: 1.00, Disparate Impact: 0.8123
```

1.1b. Code Output for Coarser Granularity

Paste your screenshot here.

```
  7%|▌         | 1/15 [00:12<02:53, 12.37s/it]Repair Level: 0.00, Disparate Impact: 0.8350
 13%|▌         | 2/15 [00:21<02:13, 10.23s/it]Repair Level: 0.01, Disparate Impact: 0.8333
 20%|▌         | 3/15 [00:31<02:04, 10.36s/it]Repair Level: 0.02, Disparate Impact: 0.8353
 27%|▌         | 4/15 [00:40<01:48,  9.88s/it]Repair Level: 0.03, Disparate Impact: 0.8438
 33%|▌         | 5/15 [00:49<01:36,  9.61s/it]Repair Level: 0.04, Disparate Impact: 0.8379
 40%|▌         | 6/15 [01:00<01:28,  9.82s/it]Repair Level: 0.06, Disparate Impact: 0.8334
 47%|▌         | 7/15 [01:08<01:14,  9.36s/it]Repair Level: 0.07, Disparate Impact: 0.8262
 53%|▌         | 8/15 [01:18<01:06,  9.47s/it]Repair Level: 0.08, Disparate Impact: 0.8464
 60%|▌         | 9/15 [01:27<00:57,  9.53s/it]Repair Level: 0.09, Disparate Impact: 0.8356
 67%|▌         | 10/15 [01:36<00:45,  9.17s/it]Repair Level: 0.10, Disparate Impact: 0.8421
 73%|▌         | 11/15 [01:46<00:37,  9.39s/it]Repair Level: 0.20, Disparate Impact: 0.8495
 80%|▌         | 12/15 [01:55<00:28,  9.39s/it]Repair Level: 0.40, Disparate Impact: 0.8683
 87%|▌         | 13/15 [02:03<00:18,  9.09s/it]Repair Level: 0.60, Disparate Impact: 0.7952
 93%|▌         | 14/15 [02:14<00:09,  9.40s/it]Repair Level: 0.80, Disparate Impact: 0.8349
100%|▌         | 15/15 [02:21<00:00,  9.42s/it]Repair Level: 1.00, Disparate Impact: 0.8123
```

1.2. Code Output for Best Repair Level and Disparate Impact at Best Repair Level

Paste your screenshot here. (Choose the single best out of all repair levels for finer/coarser. This will be the DI closest to 1.0.)

Finer Granularity:

```
Best Repair Level: 0.49
Disparate Impact at Best Level: 0.8659
```

Coarser Granularity:

```
Best Repair Level: 0.40
Disparate Impact at Best Level: 0.8683
```

1.3. Code Output for Light GBM Performance Metrics and Fairness Metrics

Paste your screenshot(s) here.

Finer Granularity:

```
Performance Metrics on Test Data (DIR Applied):
Accuracy: 0.8103
Precision: 0.6258
Recall: 0.5415
F1-Score: 0.5806
ROC-AUC: 0.7950

Fairness Metrics on Test Data (DIR Applied):
Statistical Parity Difference (SPD): -0.1097
Disparate Impact (DI): 0.8659
Equal Opportunity Difference (EOD): -0.0226
Average Odds Difference (AOD): -0.0694
Differential Fairness Bias Amplification (DFBA): 0.1727
```

```
Comparison of Performance Metrics Across Models:
Metric          Original        Reweighted      DIR Applied
------------------------------------------------------------
Accuracy        0.8176          0.8057          0.8103
Precision       0.6488          0.6140          0.6258
Recall          0.5404          0.5360          0.5415
F1-Score        0.5897          0.5724          0.5806
ROC-AUC         0.7991          0.7908          0.7950

Comparison of Fairness Metrics Across Models:
Metric          Original        Reweighted      DIR Applied
------------------------------------------------------------
SPD             -0.1385         -0.0541         -0.1097
DI              0.8339          0.9326          0.8659
SEDF            0.9115          0.7310          0.9115
```

Coarser Granularity:

```
Performance Metrics on Test Data (DIR Applied):
Accuracy: 0.8109
Precision: 0.6281
Recall: 0.5404
F1-Score: 0.5810
ROC-AUC: 0.7949

Fairness Metrics on Test Data (DIR Applied):
Statistical Parity Difference (SPD): -0.1078
Disparate Impact (DI): 0.8683
Equal Opportunity Difference (EOD): -0.0170
Average Odds Difference (AOD): -0.0699
Differential Fairness Bias Amplification (DFBA): 0.1588
```

```
Comparison of Performance Metrics Across Models:
Metric          Original        Reweighted      DIR Applied
------------------------------------------------------------
Accuracy        0.8176          0.8057          0.8109
Precision       0.6488          0.6140          0.6281
Recall          0.5404          0.5360          0.5404
F1-Score        0.5897          0.5724          0.5810
ROC-AUC         0.7991          0.7908          0.7949

Comparison of Fairness Metrics Across Models:
Metric          Original        Reweighted      DIR Applied
------------------------------------------------------------
SPD             -0.1385         -0.0541         -0.1078
DI              0.8339          0.9326          0.8683
SEDF            0.9115          0.7310          0.9115
```

***Comparison of Performance Metrics from the Original and Reweighted Models (for your convenience from the lab)**

```
Metric          Original      Reweighted
---------------------------------------
Accuracy        0.8176        0.8057
Precision       0.6488        0.6140
Recall          0.5404        0.5360
F1-Score        0.5897        0.5724
ROC-AUC         0.7991        0.7908
```

***Comparison of Fairness Metrics from the Original and Reweighted Models (for your convenience from the lab)**

Metric	Original	Reweighted
SPD	-0.1385	-0.0541
DI	0.8339	0.9326
SEDF	0.9115	0.7310

Part 2: Short-Response Questions

Question:

2.1. How does the Disparate Impact (DI) change with finer granularity? Coarser granularity?

Your response:

Using finer granularity for the Disparate Impact Remover (DIR) allows the repair levels to be evaluated in smaller increments, providing a more detailed view of how the Disparate Impact (DI) changes as the level of repair increases. In this experiment, DI gradually increased as the repair level rose, eventually reaching its highest value of 0.8659 at a repair level of 0.49. This indicates that fairness improved progressively as more repair was applied to the data. Because the repair levels were more finely spaced, it was possible to observe subtle changes in DI across many intermediate points rather than large jumps between levels. This made it easier to identify the repair level that produced the best fairness outcome and provided a clearer understanding of how fairness responds to incremental adjustments in the dataset.

With coarser granularity, the DI changes in larger, less detailed steps because the repair levels are evaluated at wider intervals. In this experiment, DI fluctuates across the tested repair levels and reaches its highest value of 0.8683 at a repair level of 0.40. After this point, DI decreases as the repair level increases further, indicating that applying too much repair can reduce the fairness benefit. Since fewer repair levels are tested, coarser granularity provides a more general overview of how DI responds to the repair process but does not capture smaller incremental changes. As a result, it may overlook more precise repair levels that could produce slightly better fairness outcomes.

Finer granularity would generally be better for the DIR. Using smaller increments for the repair levels allows the model to evaluate more intermediate points, making it easier to observe how fairness metrics change as the level of repair increases. This provides a more detailed view of the relationship between repair level and DI, helping identify the repair level that produces the best fairness outcome.

Question:

2.2. Based on your findings, which model would you recommend (Original, Reweighted, or DIR)? Justify your choice by balancing performance and fairness.

Your response:

Based on the results, the Disparate Impact Remover (DIR) model with finer granularity would be the recommended choice because it provides the best balance between model performance and fairness improvements. The DIR model maintains performance metrics that are very close to the original model, with an accuracy of 0.8103 compared to 0.8176 for the original and ROC-AUC of 0.7950 compared to 0.7991. Other performance metrics such as recall and F1-score also remain comparable, indicating that the predictive capability of the model is largely preserved.

From a fairness perspective, the DIR model improves fairness over the original model by increasing Disparate Impact from 0.8339 to 0.8659 and reducing the magnitude of Statistical Parity Difference from -0.1385 to -0.1097. Although the reweighted model achieves the best fairness metrics (DI = 0.9326 and SPD = -0.0541), it comes with a larger drop in predictive performance. Therefore, the DIR model with finer granularity represents a more balanced compromise, improving fairness while maintaining performance close to the baseline model.

Question:

2.3. Consider a scenario where you are responsible for deploying a machine learning model that shows a fairness-accuracy trade-off.

How would you approach deciding whether to prioritize fairness or accuracy? Which factors (organizational values, societal impact, or application context) would guide your decision? Reflect on how ethical dilemmas in machine learning require balancing competing values and their implications.

Your response:

When deploying a machine learning model with a fairness–accuracy trade-off, the decision I would chose would depend on the application context and the potential impact on individuals and society. If the model is used in high-stakes domains such as hiring, lending, healthcare, or criminal justice, prioritizing fairness becomes especially important because biased predictions can reinforce systemic inequalities and lead to discriminatory outcomes. For example, legal considerations, such as anti-discrimination laws and fairness standards, require reducing bias before deployment. In these cases, even a small loss in predictive accuracy may be acceptable if it significantly reduces unfair treatment of protected groups. However, in lower-risk applications such as product recommendations or ad targeting, I would prioritize accuracy and system performance because the consequences of bias are less severe.